

Briefing Document: The AI Executive Order and the Future of National Cyber Defense

Executive Summary

The recently issued Executive Order, "Promoting Advanced Artificial Intelligence Innovation and Security," marks a pivotal shift in United States policy, positioning advanced artificial intelligence at the center of national security, critical infrastructure resilience, and strategic competition. The core objective is to move beyond theoretical risk study toward the rapid operationalization of AI-enabled defenses. Key takeaways from the current landscape include:

- A Temporal Inflection Point:** AI frontier models are accelerating cyber defense capabilities, such as software red teaming, by orders of magnitude—compressing years of work into weeks.
- The "Innovation Advantage" Window:** While the United States currently maintains a head start in AI innovation, this lead is not permanent. There is an urgent need to harden domestic systems before adversaries achieve parity.
- Operationalization vs. Access:** Providing advanced AI tools to critical infrastructure operators is insufficient; the primary challenge lies in scaling and democratizing these tools for under-resourced entities like water treatment plants and electric utilities.
- New Performance Metrics:** The standard for effective cybersecurity is shifting toward detection and response times measured in single-digit minutes.
- Shift to Operational Collaboration:** Policy is moving away from simple information sharing toward deep, operational collaboration between government and industry.

Strategic Implications of the AI Executive Order

The Executive Order serves as a "North Star," signaling that AI and cybersecurity are now inseparable components of national security. It emphasizes a dual-track approach: maintaining the United States' innovation advantage while simultaneously hardening the nation's systems against AI-enhanced threats.

Critical Focus Areas

- Strategic Competition:** Utilizing AI to maintain a lead over global adversaries.
- Infrastructure Resilience:** Integrating AI-enabled defense into the sectors that support the nation's daily functions.
- Operationalized Defense:** Moving from policy frameworks to the active deployment of AI tools in defensive postures.

The AI Inflection Point: Speed and Scale

The integration of frontier models into cybersecurity has created an "inflection point" where defensive capabilities are no longer seeing incremental improvements, but exponential leaps in efficiency.

Metric	Traditional Methods	AI-Enabled (Mythos Model)
Red Teaming Code Base	1 to 2 Years	3 Weeks
Improvement Scale	5% – 15%	Major Paradigm Shift

This acceleration allows organizations to identify and remediate vulnerabilities at a pace previously considered impossible, fundamentally altering the "race" between attackers and defenders.

Challenges in Democratizing AI Defense

A significant barrier to national resilience is the disparity in resources among critical infrastructure operators. Access to frontier models (such as GPT-5.5 or specialized tools like Mythos) does not automatically translate to security for smaller organizations.

Barriers to Implementation

- **Operational Realities:** Average utilities (water, electricity) often lack the organizational structure to utilize raw AI model access effectively.
- **Resource Constraints:** Under-resourced operators require more than just software; they need scalable, democratized tools that fit their specific operational contexts.
- **The Implementation Gap:** There is a critical need for an "AI cybersecurity clearinghouse" and voluntary frameworks to help these entities bridge the gap between having a tool and knowing how to use it for defense.

Redefining Cyber Defense Metrics

The evolution of AI requires a corresponding evolution in how the success of cybersecurity is measured. The transition from human-led to AI-augmented defense necessitates a focus on extreme speed and collaboration.

Key Operational Shifts

- **Response Time:** Detection and response must now be measured in **single-digit minutes**. Times measured in days, weeks, or "never" are no longer acceptable in an AI-accelerated threat environment.
- **Beyond Information Sharing:** The document emphasizes that "information sharing alone" is insufficient. The goal is "operational collaboration," where industry and government work in tandem to manage the surge in vulnerabilities and improve system hardening.

Essential Quotes

In three weeks, Mythos was able to conduct one to two years' worth of red teaming on our own code base. We're not talking 5 percent better, 10 percent better, 15 percent better. We're talking about doing something in three weeks that would have taken us one, if not more, years previously. So that is an inflection point." — **Daniel Kroese**"We have a head start, but it is not an infinite head start." — **Daniel Kroese**"We also have to recognize that for your average electric utility or water treatment plant, if we were to give them Mythos or GPT-5.5 access tomorrow, due to the operational realities of how they are organized, they wouldn't know what to do with it. So it's not as simple as just flicking on access. It's about how do we scale and democratize the Cyber defense benefits of these models." — **Daniel Kroese**"Detection response times must be measured in single-digit minutes, not days, weeks, or never." — **Daniel Kroese**"This isn't about information sharing alone. It's about operational collaboration." — **Daniel Kroese**